

L1 R 008secs MED
00:02:14 VEH:007

Fig. 4: Display on the LCD

tion of respective relays.

The system information is displayed on the LCD. Port pins P1.0 through P1.7 of the microcontroller are connected to data port pins D0 through D7 of the LCD. Control pins register-select (RS), read / write (R/W) and enable (E) are connected to port pins P2.7, P2.6 and P2.5, respectively. All the data is sent to the LCD in ASCII format and the commands in hex format. Preset VR1 connected to pin 3 of the LCD allows contrast control.

Fig. 4 shows a traffic profile displayed on the LCD. L1 in the first line of the LCD indicates the lane or location where this system has been installed. The next character (R, Y or G for red, yellow or green) indicates red as the currently-running traffic light. The next three digits indicate the remaining time (in seconds) of the current traffic light. The next three characters (LOW, MED and HIGH for low, medium and high) show the current traffic profile based on the number of vehicles that passed through the IR transmitter-receiver pairs

in the last controlling interval.

The second line of the LCD starts with the system running time in hours, minutes and seconds. Next is the number of vehicles that have passed through the traffic light's IR sensor system. Based on this vehicle count and traffic profile, the system

applies traffic delays for the next controlling interval.

The IR sensor system is installed before the traffic light, say, 500 metres away from the traffic light. Applying the traffic profile and respective traffic delays based on the previous controlling interval helps the system to manage traffic congestion to some extent.

Switches 'Up,' 'Down' and 'Enter' connected to pins P2.0, P2.1 and P2.2 of the microcontroller, respectively, allow the user to change the traffic profile, vehicle count, traffic delays in respective profiles and controlling interval. 'Up' switch is for increasing and 'Down' switch is for decreasing the default values of system parameters. The default parameter values for different traffic profiles and parameter ranges are shown in Tables I and II, respectively. The value set by the user using 'Up' and 'Down' switches is accepted only after pressing 'Enter' switch. The system can be made to enter configuration state by pressing 'Enter' key while running.

Software

The source program for the microcontroller-based intelligent traffic light system is written in Assembly language

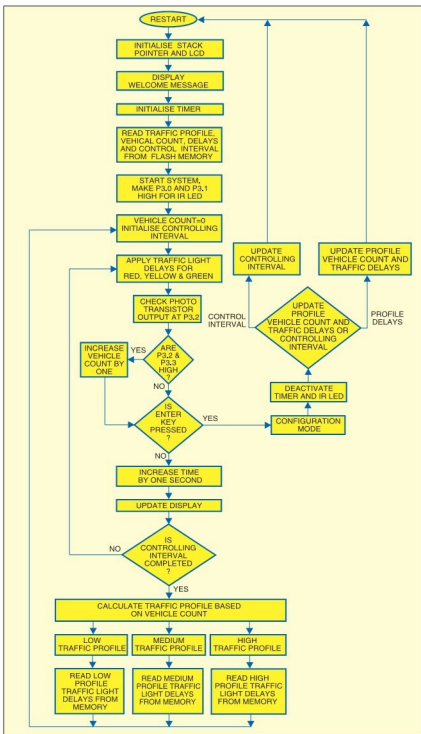


Fig. 5: Software flow-chart